

Section 1E. Expectation and Variance

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Random Variables: Expectation

- ▶ Given a discrete r.v. X with PMF p_X and a function $g : \mathbb{R} \rightarrow \mathbb{R}$, the **expectation** (or **expected value**) of $g(X)$ is defined as

$$\mathbb{E}[g(X)] = \sum_{x \in \mathcal{X}} g(x) p_X(x)$$

where $\mathcal{X}$ is the set of possible values X may take

- ▶ If X is continuous with PDF f_X , the **expected value** of $g(X)$ is

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

When $g(x) = x$, we have that $\mathbb{E}[g(X)] = \mathbb{E}[X]$, which is known as the **mean** of X .

- ▶ **Properties** of the expectation:
 - ▶ For any constant $a \in \mathbb{R}$, $\mathbb{E}[a] = a$ and $\mathbb{E}[ag(X)] = a\mathbb{E}[g(X)]$
 - ▶ $\mathbb{E}[f(X) + g(X)] = \mathbb{E}[f(X)] + \mathbb{E}[g(X)]$

Random Variables: Variance

- ▶ The **variance** of a r.v. X is defined as

$$\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

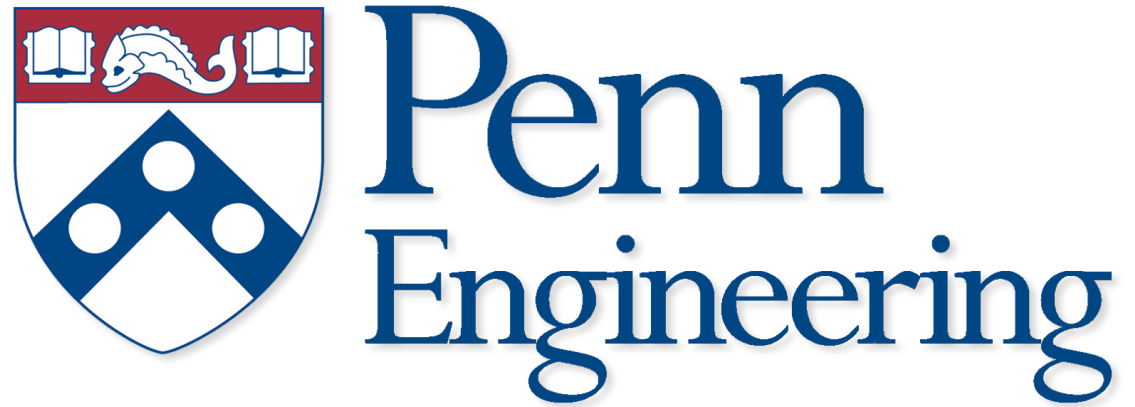
- ▶ **Properties** of the variance:

1. For any constant $a \in \mathbb{R}$, $\text{Var}[ag(X)] = a^2 \text{Var}[g(X)]$
2. Given two independent r.v.'s X and Y , we have that $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$

- ▶ **Example:** Calculate the mean and variance of a r.v. with a uniform PDF, i.e., $f_X(x) = 1$ for all $x \in [0, 1]$, 0 elsewhere.

$$\mathbb{E}[X] = \int_0^1 x \, dx = 1/2 \text{ and } \mathbb{E}[X^2] = \int_0^1 x^2 \, dx = 1/3$$

$$\text{Hence, } \text{Var}[X] = 1/3 - 1/4 = 1/12$$



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